Welcome to 151B: Deep Learning

Instructor: Annapurna Vadaparty

TA: Ben Bao

Tutors: Alexander Radulescu, Trevor Duong

What is something you'd like to model, and what would the metrics be?

- Yesterday we heard Jay talk about similarities to the way humans make **mistakes** and the amount of **time it takes to do a task**, and how these are the metrics we use to assess whether a model is doing what humans are doing
- What is something you're interested in modeling, and what would be relevant metrics for it?
- Might be loss/accuracy, might be something else. Even if it is "build the most accurate model", what accuracy are you measuring? E.g. for something like economic modeling, does it take externalities into account?
- Chat with the person next to you!

Externality

46 languages

Article Talk Read Edit View history

From Wikipedia, the free encyclopedia

This article **may incorporate text from a large language model, which is prohibited in Wikipedia articles**. It may include hallucinated information, copyright violations, claims not verified in cited sources, original research, or fictitious references. Any such material should be removed. *(October 2025)* *(Learn how and when to remove this message)*

In economics, an **externality** is a cost or benefit to an uninvolved third party that arises as an effect of another party's (or parties') activity. Many externalities can be considered as unpriced components that are involved in either consumer or producer consumption. Air pollution from motor vehicles is one example. The cost of air pollution to society is not paid by either the producers or users of motorized transport. Water pollution from mills and factories are another example. All (water) consumers are made worse off by pollution but are not compensated by the market for this damage.

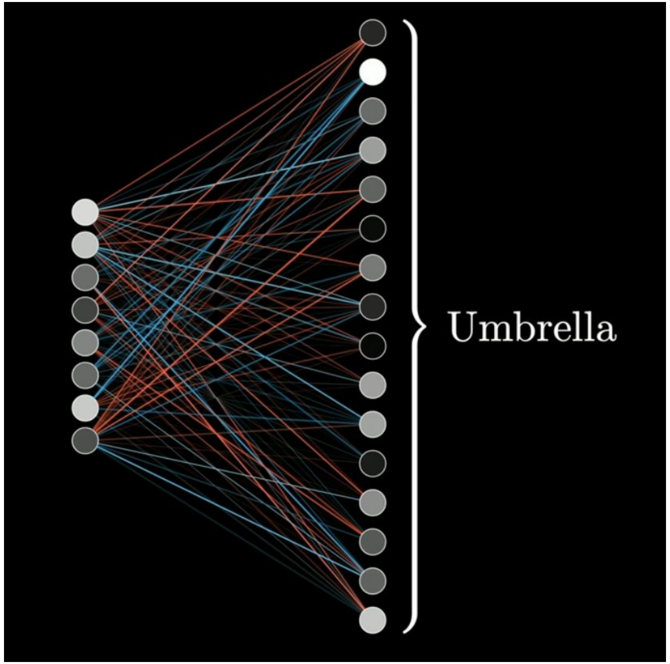
Air pollution from motor vehicles is an example of a negative externality. The costs of the air pollution for the rest of society is not compensated for by either the producers or users of motorized transport.

The concept of externality was first developed by Alfred Marshall in the 1890s[1] and achieved broader attention in the works of economist Arthur Pigou in the 1920s.[2] The prototypical example of a negative externality is environmental pollution. Pigou argued that a tax, equal to the marginal damage or marginal external cost, (later called a "Pigouvian tax") on negative externalities could be used to reduce their incidence to an efficient level.[2] Subsequent thinkers have debated whether it is preferable to tax or to regulate negative externalities,[3] the optimally efficient level of the Pigouvian taxation,[4] and what factors cause or exacerbate negative externalities, such as providing investors in corporations with limited liability for harms committed by the corporation.[5][6][7]

Other thoughts/reflections from Jay’s talk?

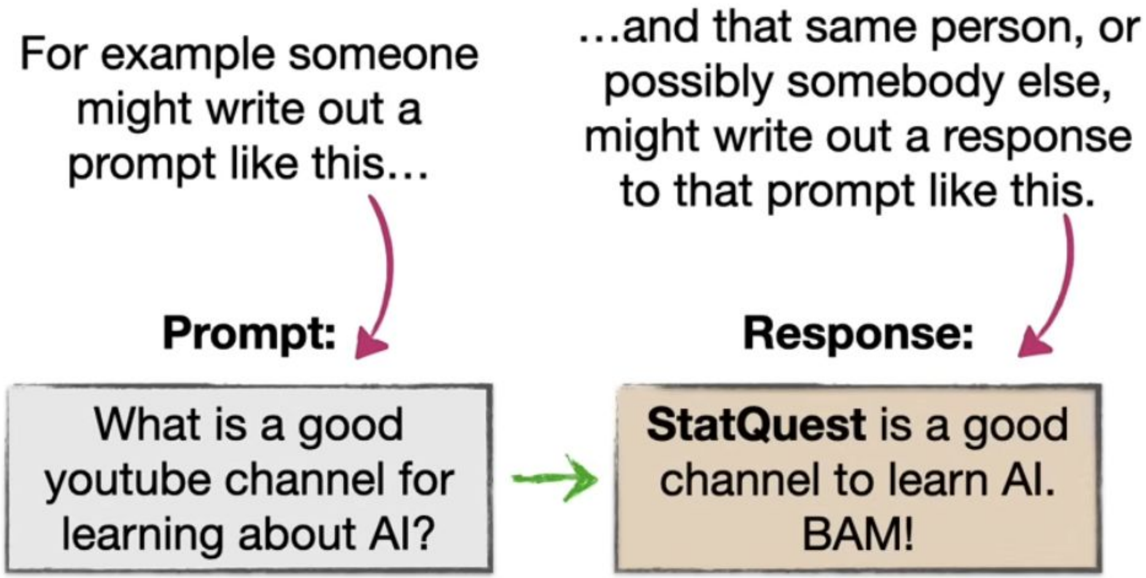
Superposition

- What is superposition in LLMs?
- Can't have every single idea represented in its own vector–too many!
- Similar to brains…we don't have a different neuron for every idea
  - Population codes!



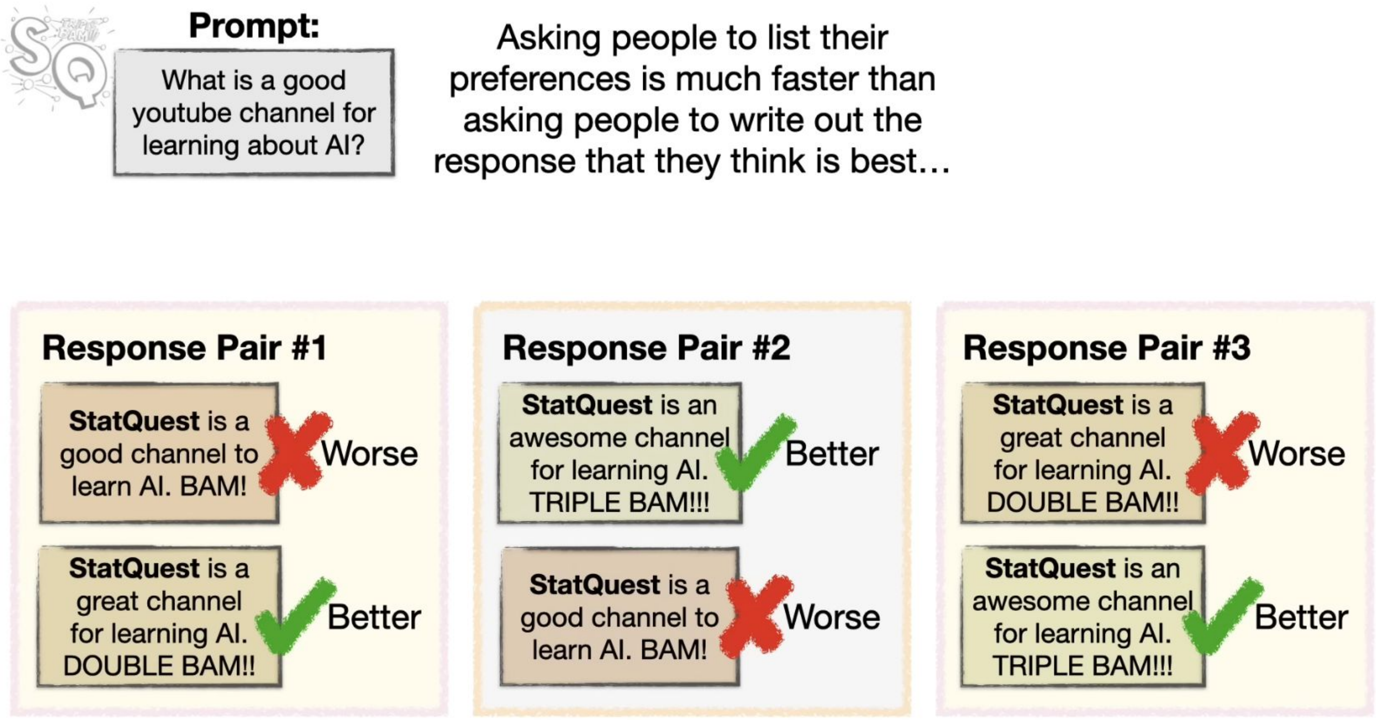
3b1b, How might LLMs store facts | Deep Learning Chapter 7

Supervised Fine Tuning



StatQuest, Reinforcement Learning with Human Feedback (RLHF), Clearly Explained!!!

RLHF



StatQuest, Reinforcement Learning with Human Feedback (RLHF), Clearly Explained!!!

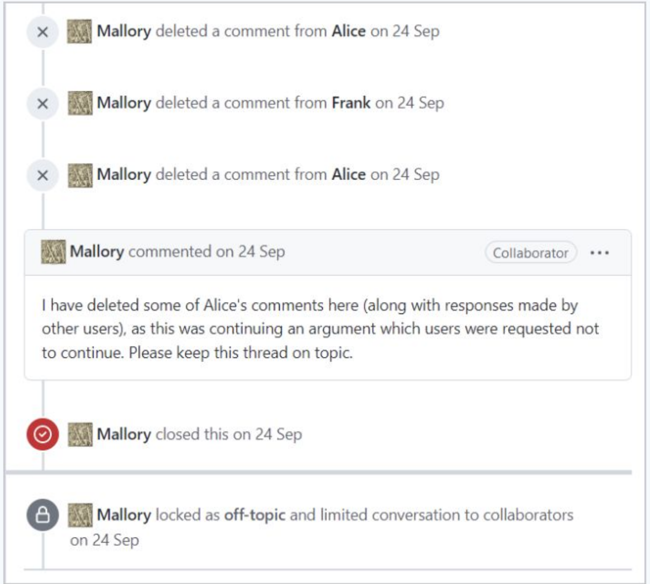
What is content moderation?

- What is content moderation? What tools/contexts have you heard about it in?
- Chat with the person next to you

Content moderation

From Wikipedia, the free encyclopedia

Content moderation, in the context of websites that facilitate user-generated content, is the systematic process of identifying, reducing,[1] or removing user contributions that are irrelevant, obscene, illegal, harmful, or insulting. This process may involve either direct removal of problematic content or the application of warning labels to flagged material. As an alternative approach, platforms may enable users to independently block and filter content based on their preferences.[2] This practice operates within the broader domain of trust and safety frameworks.



Comment moderation on a GitHub discussion, where a user called Mallory has deleted several comments before closing the discussion and locking it

Various types of Internet sites permit user-generated content such as posts, comments, videos including Internet forums, blogs, and news sites powered by scripts such as phpBB, a wiki, PHP-

How does content moderation work? Who does it?

- What do you think?

Working conditions [edit]

Employees work by viewing, assessing and deleting disturbing content.[11] *Wired* reported in 2014, they may suffer psychological damage.[12][13][14][5][15] In 2017, the Guardian reported secondary trauma may arise, with symptoms similar to PTSD.[16] Some large companies such as Facebook offer psychological support[16] and increasingly rely on the use of artificial intelligence to sort out the most graphic and inappropriate content, but critics claim that it is insufficient.[17] In 2019, NPR called it a job hazard.[18] Non-disclosure agreements are the norm when content moderators are hired. This makes moderators more hesitant to speak up about working conditions or organize.[7]

Psychological hazards including stress and post-traumatic stress disorder, combined with the precarity of algorithmic management and low wages make content moderation extremely challenging.[19][20]:123 The number of tasks completed, for example labeling content as copyright violation, deleting a post containing hate-speech or reviewing graphic content are quantified for performance and quality assurance.[7]

A tale as old as the Internet. Is anything unique to GenAI?

- Not a lot, but still worth mentioning
- One difference: generation of disturbing content
- Classic moderation is reactive gatekeeping — leave up / take down
- RLHF/Supervised Fine Tuning safety work often requires the worker to author the harmful content: red-teaming, adversarial prompt generation, writing the disturbing completion so the model learns the boundary
- (MTurk-era crowdwork also sometimes had similar tasks, but not to this scale)

“Sacrifice Zones”

Karen Hao & Hasan Minhaj, discussing “Empire of AI”

Another externality: data centers

The New York Times

India Is Moving Fast to Catch Up in A.I. A Coastal City Fears the Fallout.

With India lagging in the technology, officials are embracing giant data centers. But critics say the megaprojects will use up energy and water, without providing long-term jobs.

Another externality: data centers

But in Visakhapatnam, many residents are skeptical of the benefits.

Hyperscale data centers guzzle energy and water, and are not big job creators. These issues, which are particularly acute in India, have already spurred opposition in the United States, where data center projects worth tens of billions of dollars have been blocked or delayed by opposition from local communities.

The hasty pace of the projects in Andhra Pradesh has some residents, activists and former officials concerned that tech giants are coming here to exploit the Indian government's desperate need for a place in the A.I. boom.

India is offering large subsidies to Google and its local partner, the Adani Group — from discounted land, energy and water, to extensive tax breaks — for the kind of project that communities in the U.S. are delaying.

What do we do now?

- Local LLMs, as much as possible
    - No further training and RLHF needed
    - No further API calls needed
    - Similar logic to “reduce, reuse, recycle”
- Be involved in local government–be aware of what your city is doing
    - Whether that’s here or back home! Vote in whatever place you have the power to vote in.
- Thoughts? Concerns?

Project Updates

Tracks

Communication Track

Create a video, blog post, article, game, or anything else that explains or teaches some idea in deep learning in a novel context. You are not required to formally evaluate how your artifact landed with an audience, though you're welcome to.

Deliverables:

- Your artifact
- A short paper explaining your design decisions and creation process, written as an experience report
- A 10–15 minute presentation

More about experience reports: SIGCSE 2026 Papers track description.

Model-Building Track

Build a solution using any deep learning algorithms and architectures to score well on the ARC multiple choice science question dataset.

Deliverables:

- Your code
- The related works, methods, and results section of a research paper
- A 3–5 minute lightning talk

More about research papers: SIGCSE 2026 Papers track description.

▾ Week 5: Final Project Report and Presentation

Some terms to prepare for Prof. Gary Cottrell's talk

- Gabor filters
- Fovea
- Log-polar transform in V1 (cortical magnification)

Gabor Filters

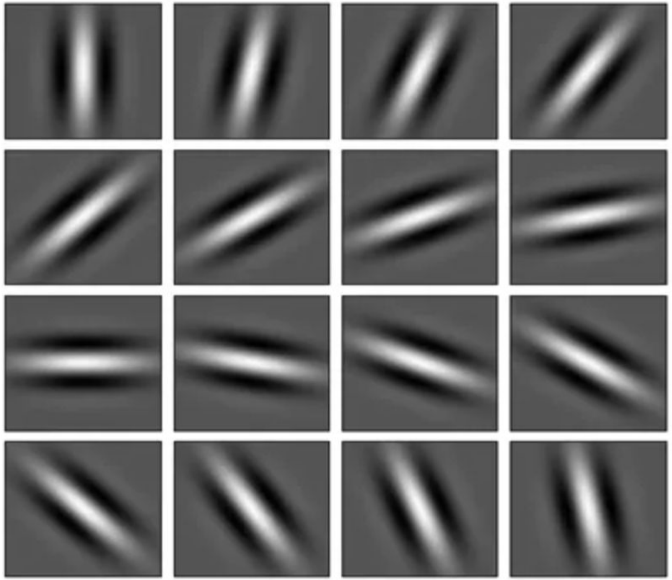
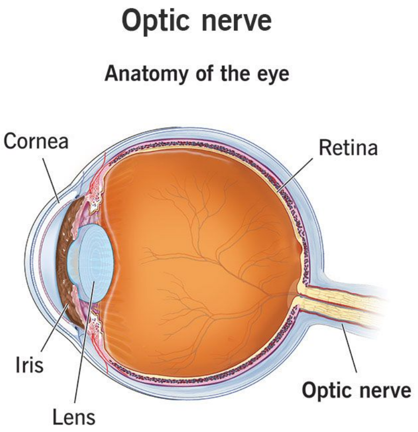
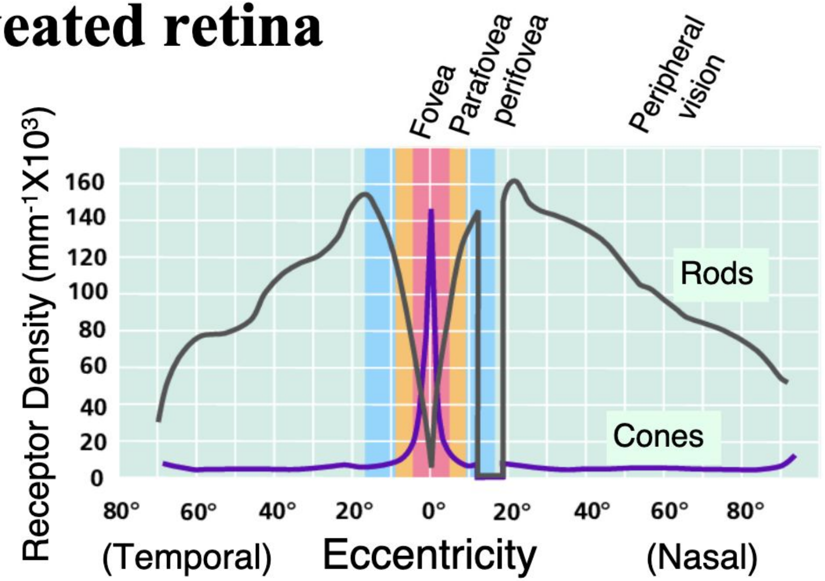


Figure 2: A bank of 16 Gabor filters oriented at an angle of 11.250 (i.e. if the first filter is at 00, then the second will be at 11.250, the third will be at 22.50, and so on.)

https://medium.com/@anuj_shah/through-the-eyes-of-gabor-filter-17d1fdb3ac97

Anatomical Constraints:
The foveated retina

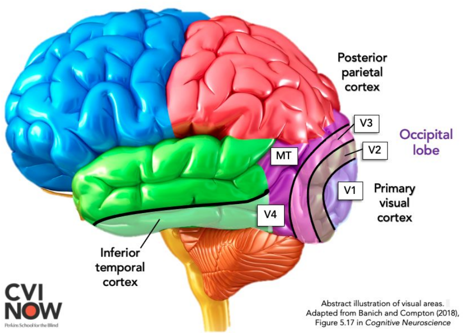


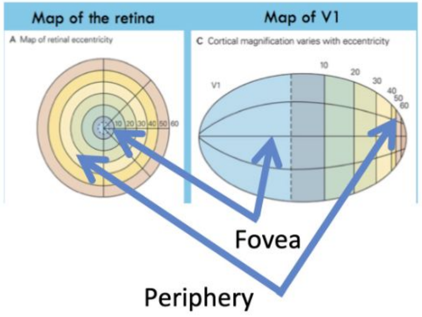


The fovea has a high density of photoreceptors, which drop off in the periphery

As a result, we move our eyes about 172,000 times a day

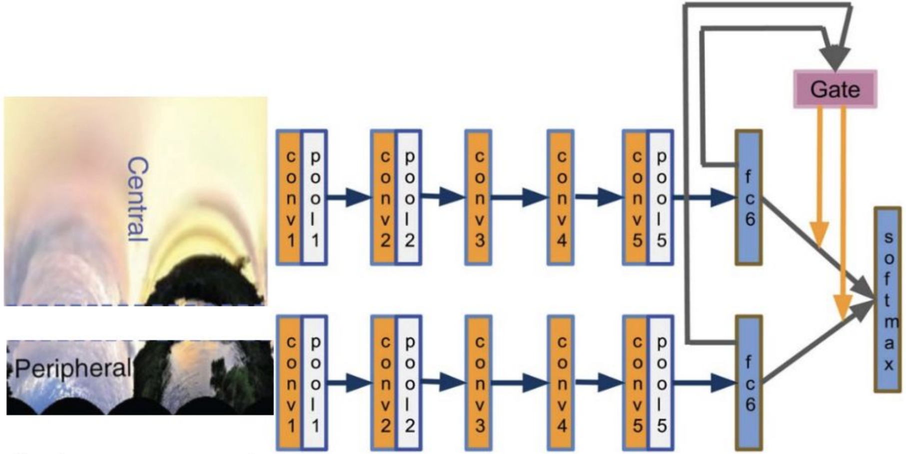
Anatomical Constraints: The log-polar transform





- There is a log-polar transform from the visual field to V1
- This separates central from peripheral representations
- The result is a huge “cortical magnification” of the central visual field representation - and a corresponding shrinkage of peripheral vision
- (This is not *really* magnification: It just results from there being more receptors in the fovea, fewer in the periphery)

The Model 2.0: Anatomical Constraints



- For what comes next:
  - We used three models pre-trained on imagenet (AlexNet, VGG, and GoogLeNet)
  - And then fine-tuned them on foveated images (The Model 1.9)
  - And fine-tuned them on log-polarized images (The Model 2.0)
  - On versions of a scene classification dataset

"I find the hoops CV people jump through to achieve this (image pyramids, three networks at different scales) rather baroque…"

Prof. Cottrell, The Model 2.0